

Dhairyakumar Shah

shahdhairya1380@gmail.com | +44 7901 170839 | London, UK | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL PROFILE

AI / Machine Learning developer with an MSc in Data Science (Distinction) and hands-on experience building and deploying agentic, LLM-powered applications. Designed and shipped a Retrieval-Augmented Generation (RAG) chatbot using Llama, LangChain and a vector database, deployed to the cloud for real users. Strong in Python and the modern ML stack, with a solid grounding in data pipelines, validation and stakeholder reporting, developed across both retail trading analysis and healthcare data operations. Combines a rigorous, physics-trained analytical mindset with practical engineering skills to build reliable, production-ready AI systems.

KEY SKILLS & TECHNICAL COMPETENCIES

Agentic AI & LLMs: RAG pipelines, LangChain, agent frameworks, prompt engineering, Llama / open-source LLMs, LlamaIndex

ML & Data: Python (pandas, NumPy, scikit-learn), R, NLP, model evaluation, feature engineering

Palantir Stack: Palantir Foundry (Pipeline Builder, Ontology Manager), Palantir AIP Analyst, ontology modelling, action type configuration

Data & Retrieval: Vector databases (FAISS / Chroma / Pinecone), SQL, data pipeline integrity & validation, embeddings

Deployment & Tooling: Cloud (AWS / Azure / GCP), Railway, Git, Jupyter, REST APIs, Microsoft Teams

Professional: Stakeholder reporting, clear documentation, problem-solving under pressure, flexible incl. out-of-hours

KEY PROJECTS

Agentic Medical Information Chatbot (RAG) | Personal Applied Project | [Website](#)

- Designed and built an agentic chatbot that answers medical and health-information questions using a Retrieval-Augmented Generation (RAG) pipeline, grounding responses in a curated knowledge base to reduce hallucination.
- Used an open-source Llama model with LangChain to orchestrate retrieval, prompt construction and response generation, and a vector database to store and semantically search document embeddings.
- Implemented document ingestion, chunking and embedding, with retrieval tuned for relevance, and structured the agent to call tools and follow multi-step reasoning.
- Deployed the application to the cloud via Railway, making it accessible to real users, and iterated on performance and answer quality based on usage.

Palantir AIP — NHS A&E Operational Intelligence Workflow | Technical Assessment | [YouTube Link](#)

- Built an end-to-end AI workflow on Palantir AIP using real NHS England A&E open data (OGL v3.0) spanning 2,372 records across 200+ trusts for the 2024–25 period, independently designed and delivered from scratch.
- Engineered a Foundry batch pipeline and modelled an AE_Attendance ontology to structure and expose trust-level performance metrics — breach rates, 12-hour waits, total attendances — for downstream AI querying.
- Configured Palantir AIP Analyst to enable natural language operational querying, replacing the need for manual SQL analysis and ad-hoc reporting.
- Surfaced key insights: identified East Cheshire NHS Trust as the highest breach-rate trust in March 2025 (49.96%) and NHS England Midlands as the region with the most 12-hour waits across 2024–25 (123,264 cases).
- Configured an operational action type to flag high-risk trusts for escalation, demonstrating a full agentic workflow from data ingestion to decision support.
- Stack:** Palantir Foundry, Palantir AIP Analyst, Python, NHS England Open Data

PROFESSIONAL EXPERIENCE

Customer & Trading Manager | Sainsbury's, London (FTSE 100)

Apr 2023 – Present

- Built and maintained automated weekly KPI trackers in Excel, synthesising performance metrics across the department and delivering regular data-driven reports to senior management.
- Analysed daily and weekly trading data against targets to identify underperformance and recommend corrective action, applying strong analytical and critical-thinking skills under pressure.
- Identified demand patterns through historical data analysis to support proactive planning and forecasting.
- Drove daily systematic validation and error correction across stock and availability data, supporting 97.4% stock availability and a 10% reduction in ordering errors.
- Made prompt operational decisions from real-time data and upheld full compliance with internal audit and governance frameworks in a highly regulated environment.

Process Associate | ifedora Healthcare Solutions, Ahmedabad, India*May 2021 – Oct 2022*

- Processed and validated patient billing documentation and prescription records for US medical practices within a revenue cycle management operation certified to ISO 9001:2015 and ISO/IEC 27001:2022.
- Reviewed treatment information by condition and cross-checked it against source records, identifying discrepancies, missing information and inconsistencies and correcting errors at source to protect data integrity before downstream billing.
- Maintained consistent accuracy standards across high volumes of patient-identifiable records, working to strict confidentiality, information security and data protection procedures.
- Entered, amended and maintained data in billing and record-keeping systems, gaining end-to-end understanding of healthcare administrative processes from referral through to billing.
- Built working fluency in clinical and pharmaceutical terminology, including medication names and standard treatment and procedure codes across multiple specialities.
- Liaised with US healthcare practices by telephone to resolve missing or inconsistent information, escalating queries outside agreed parameters to senior colleagues.

EDUCATION

MSc Data Science with Advanced Research — Distinction | University of Hertfordshire, UK *2023 – 2025***MSc Physics — First Class | Gujarat University, India** *2019 – 2021***BSc Physics — Second Class** **| Gujarat University, India**
*2016 – 2019***CERTIFICATIONS & ADDITIONAL INFORMATION**

- Ongoing CPD — agentic AI, LLMs, applied machine learning; Udemy: Complete Data Science, Machine Learning, DL & NLP Bootcamp (99+ hours).
- Basic DBS held (comfortable obtaining Enhanced DBS); Network Storage Certificate; Field Technician Certificate.
- English: Fluent (UK MSc satisfies B2 Skilled Worker requirement). Availability: Full-time, flexible to work extended hours. Location: London.